THE COTANGENT COMPLEX OF A TOR INDEPENDENT FIBRE PRODUCT OF ALGEBRAIC SPACES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file cotangent.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The statement refers to the situation and the fibre product equation fixed earlier in the chapter, and displays the resulting decomposition of the cotangent complex.

Editorial note. The five references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the retained preamble of this chapter (sources/stacks-cotangent/retained-preamble.tex, lines 1-63), containing the Xy-pic setup, the comment environments, the theorem declarations and the macros including \NL.

Allowed standard prerequisites (editorial).

- The comparison between the cotangent complexes of algebraic spaces and of their scheme presentations. Lemma lemma-compare-spaces-schemes, used at proof lines 4812-4814. Reduces the vanishing statement to schemes.
- Tor independence for derived categories of algebraic spaces. Derived Categories of Spaces, Lemma spaces-perfect-lemma-tor-independent, used at proof lines 4812-4814. Transfers the Tor independence hypothesis to the presentation.
- The comparison of the cotangent complex over an affine open with the ring-level complex. Lemma lemma-morphism-affine-schemes (4129-4140, produced as L44141), used at proof lines 4815-4819. Reduces the affine scheme case to a tensor product of rings.
- The ring-level case. Lemma lemma-tensor-product-tor-independent, used at proof lines 4817-4819. Settles the final case.

Source reference. Original equation 09DK: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11868–11868. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08VF: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11233–11233. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08T3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11151–11151. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:cotangent.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 09DA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 11858–11858. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 08IQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 10823–10823. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *In the situation above, if X and Y are Tor independent over B, then the object E in (09DK) is zero. In this case we have*

$$L_{X \times_B Y/B} = Lp^* L_{X/B} \oplus Lq^* L_{Y/B}$$

Proof. Choose a scheme W and a surjective étale morphism $W \to B$. Choose a scheme U and a surjective étale morphism $U \to X \times_B W$. Choose a scheme V and a surjective étale morphism $V \to Y \times_B W$. Then $U \times_W V \to X \times_B Y$ is surjective étale too. Hence it suffices to prove that the restriction of E to $U \times_W V$ is zero. By Lemma 08VF and Derived Categories of Spaces, Lemma 08IQ this reduces us to the case of schemes. Taking suitable affine opens we reduce to the case of affine schemes. Using Lemma 08T3 we reduce to the case of a tensor product of rings, i.e., to Lemma 09DA. □